Supporting Information

AI-Driven X-ray CT Image Characterization for Accelerating Polymer Composite Design

Xiaoyang Zheng, Bin Xu, Junichiro Shiomi, Takayuki Yamada**

X.Z., B.X., J.S., T.Y.
Institute of Engineering Innovation, Graduate School of Engineering, The University of Tokyo, Tokyo 113-8656, Japan
Email Address: zheng@aoni.waseda.jp; t.yamada@mech.t.u-tokyo.ac.jp

X.Z.
Global Center for Science and Engineering, Faculty of Science and Engineering, Waseda University, Tokyo 169-8555, Japan

1 Deep Learning Implementation and Training

1.1 PocomAI-seg

PocomAI-seg, built upon the boundary-aware segmentation network [1], consists of a prediction module and a residual refinement module. The prediction module is a U-Net-like encoder–decoder network. The encoder contains an input convolutional layer followed by four stages composed of basic residual blocks adapted from ResNet-34 [2]. The decoder is approximately symmetric to the encoder, and each stage consists of three convolutional layers, each followed by batch normalization and a ReLU activation function.

To further capture global information, a bridge stage is inserted between the encoder and decoder. This stage consists of three convolutional layers with 128 dilated 3×3 filters, each followed by batch normalization and a ReLU activation function. The residual refinement module is a lightweight encoder–decoder network, in which both the encoder and decoder contain four stages, and each stage contains one convolutional layer followed by batch normalization and a ReLU activation function. The bridge stage of the refinement module also contains one convolutional layer followed by batch normalization and a ReLU activation function. Non-overlapping max pooling is used for downsampling in the encoder, and bilinear interpolation is used for upsampling in the decoder. The output of the refinement module is used as the final segmentation map. The training loss is defined as a hybrid loss:

$$\mathcal{L} = \mathcal{L}_{bce} + \mathcal{L}_{ssim} + \mathcal{L}_{iou} \tag{S1}$$

where $\mathcal{L}_{bce}$, $\mathcal{L}_{ssim}$, and $\mathcal{L}_{iou}$ denote the binary cross-entropy (BCE) loss [3], structural similarity index measure (SSIM) loss [4], and intersection-over-union (IoU) loss [5], respectively.

The BCE loss, which is widely used for binary classification and segmentation, is defined as:

$$\mathcal{L}_{bce} = -\sum \left(G \log(S) + (1 - G) \log(1 - S)\right) \tag{S2}$$

where $G \in \{0, 1\}$ is the ground-truth label of a pixel and S is the predicted probability of the segmented object.

The SSIM loss was originally developed for image-quality assessment and captures structural information in an image. Let $\mathbf{x} = \{x_j : j = 1, \ldots, N^2\}$ and $\mathbf{y} = \{y_j : j = 1, \ldots, N^2\}$ denote the pixel values of two corresponding patches of size $N \times N$ extracted from the predicted probability map S and the binary ground-truth mask G, respectively. The SSIM loss is defined as:

$$\mathcal{L}_{ssim} = 1 - \frac{(2\mu_x\mu_y + C_1)(2\sigma_{xy} + C_2)}{(\mu_x^2 + \mu_y^2 + C_1)(\sigma_x^2 + \sigma_y^2 + C_2)} \tag{S3}$$

where μ_x and μ_y are the means of $\mathbf{x}$ and $\mathbf{y}$, σ_x and σ_y are their standard deviations, and σ_{xy} is their covariance. The constants $C_1 = 0.01^2$ and $C_2 = 0.03^2$ are introduced to avoid numerical instability.

The IoU loss, which measures the overlap between the ground truth and the prediction, is defined as:

$$\mathcal{L}_{iou} = 1 - \frac{G \cap S}{G \cup S} \tag{S4}$$

PocomAI-seg takes a grayscale image of size $256 \times 256 \times 1$ as input, predicts a coarse map of size $128 \times 128 \times 1$, and subsequently outputs a refined binary map of size $256 \times 256 \times 1$. PocomAI-seg was trained using $256 \times 150{,}000$ X-ray CT-style images $I_{CT,i}$ and their corresponding segmented binary images $I_{b,i}$ from the multimodal dataset, with a training-to-validation split ratio of 0.8:0.2.

Training for 100 epochs required approximately 39 h on a single NVIDIA RTX A6000 graphics card (48 GB GPU memory) running on a Linux system. The training process used a batch size of 32 and the Adam optimizer with a learning rate of 0.0001. Figure S3 shows the evolution of the training loss and validation mean absolute error (MAE), where the loss rapidly converged to approximately 0.0025 after 20 epochs, accompanied by similar convergence of the validation MAE. These results demonstrate stable and robust training, and the small validation MAE further confirms that the trained PocomAI-seg can accurately segment X-ray CT images. Checkpoints of the trained PocomAI-seg were saved for the AI-driven framework PocomAI.

1.2 PocomAI-gen

PocomAI-gen was built based on a conditional denoising diffusion probabilistic model [6, 7]. Compared with other popular deep generative models, such as generative adversarial networks (GANs), variational autoencoders (VAEs), autoregressive models, and flow-based generative models, diffusion models are known for stable, diverse, and realistic image generation [8]. In PocomAI-gen, two processes are involved: a forward noising process q and a backward denoising process p.

The forward noising process q is defined as a Markov chain with T steps that gradually adds Gaussian noise $\mathcal{N}(0, I)$ to a voxel grid according to a variance schedule $\beta_1, \ldots, \beta_t, \ldots, \beta_T$:

$$q(x_t \mid x_{t-1}) = \mathcal{N}(x_t; \sqrt{1 - \beta_t}\, x_{t-1}, \beta_t I) \tag{S5}$$

$$q(x_{1:T} \mid x_0) = \prod_{t=1}^{T} q(x_t \mid x_{t-1}) \tag{S6}$$

where $x_1, \ldots, x_T$ are latent variables with the same dimensionality as the data $x_0 \sim q(x_0)$. A useful property of this forward process is that x_t can be sampled directly at an arbitrary timestep t using $\alpha_t := 1 - \beta_t$ and $\bar{\alpha}_t := \prod_{s=1}^{t} \alpha_s$, which gives:

$$q(x_t \mid x_0) = \mathcal{N}(x_t; \sqrt{\bar{\alpha}_t}\, x_0, (1 - \bar{\alpha}_t)I) \tag{S7}$$

During the backward denoising process p, an autoencoder is used to predict the added noise at a given timestep. The reverse process is written as:

$$p_\theta(x_{t-1} \mid x_t) = \mathcal{N}(x_{t-1}; \mu_\theta(x_t, t), \Sigma_\theta(x_t, t)) \tag{S8}$$

Here, a conditional 3D U-Net is used to predict the noise in the input voxel grid x_t at timestep t by learning the mean and variance terms of the reverse process [9]. The conditional 3D U-Net is constructed using 3D convolutional neural networks with ResNet layers, time embeddings, and attention layers. For conditional generation, a vision encoder is incorporated to encode the input images, and the resulting embeddings are concatenated with the voxel features at the beginning of the ResNet blocks in the U-Net.

The loss function of the 3D conditional diffusion model is defined as the mean squared error between the predicted noise ϵ_θ and the true noise ϵ, where $\epsilon \sim \mathcal{N}(0, I)$:

$$\mathcal{L}_{simple}(\theta) := \mathbb{E}_{t,x_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}x_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\right] \tag{S9}$$

PocomAI-gen was trained iteratively using the following procedure for each training batch: random timesteps were sampled according to the batch size; the forward diffusion process was applied to the input voxel grids to generate noised voxel grids x_t and the corresponding ground-truth noise ϵ; the U-Net then predicted ϵ_θ from the noised voxel grids x_t, the timesteps t, and the conditional image labels; and finally, the loss between ϵ_θ and ϵ was computed to update the trainable parameters of the U-Net by gradient descent.

After training, image-conditioned generation is performed as follows: a latent variable z_T is sampled from Gaussian noise $\mathcal{N}(0, I)$, and then the reverse diffusion process is repeated for timesteps $t = T, \ldots, 1$ to sample $z_{t-1} \sim p_\theta(z_{t-1} \mid z_t)$ according to:

$$z_{t-1} = \frac{1}{\sqrt{\alpha_t}}\left(z_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\epsilon_\theta(z_t, t)\right) + \sigma_t \eta \tag{S10}$$

where $\eta \sim \mathcal{N}(0, I)$, $\Sigma_\theta(z_t, t) = \sigma_t^2 I$, and ϵ_θ is the function approximator represented by the conditional 3D U-Net. During each timestep, the image embeddings are injected into the ResNet blocks of PocomAI-gen. The final 3D voxel grid z_0 is obtained after completing all timesteps.

PocomAI-gen was trained using segmented binary images I_b of size $256 \times 256 \times 1$ and their corresponding coarse voxel grids V_c of size $64 \times 64 \times 64 \times 1$ from the multimodal dataset. Training for 100 epochs required approximately 204 h on a single NVIDIA RTX A6000 graphics card (48 GB GPU memory) running on a Linux system. The training process used a batch size of 8 and the Adam optimizer with a learning rate of 0.0001. Figure S4 shows the evolution of the training loss, which rapidly converged to approximately 0.001 after 25 epochs. These results demonstrate stable and robust training. Checkpoints of the trained PocomAI-gen were saved for the AI-driven framework PocomAI.

1.3 PocomAI-sup

PocomAI-sup was built upon an enhanced deep residual network consisting of multiple residual blocks [10]. The network comprises an input convolutional layer, eight residual blocks, a convolutional layer, a convolutional upscaling layer, and an output convolutional layer. Each residual block consists of two convolutional layers with a ReLU activation function between them. Compared with the original ResNet design, batch normalization layers are removed, together with the final ReLU activation, because batch normalization constrains the feature range and may reduce output flexibility. Removing batch normalization also reduces GPU memory consumption, as these layers require memory comparable to that of the preceding convolutional layers. PocomAI-sup refines voxel grids from a size of $64 \times 64 \times 64 \times 1$ to $128 \times 128 \times 128 \times 1$.

PocomAI-sup was trained using coarse voxel grids V_c of size $64 \times 64 \times 64 \times 1$ and fine voxel grids V_f of size $128 \times 128 \times 128 \times 1$ from the multimodal dataset with a training-to-validation split ratio of 0.8:0.2. Training for 100 epochs required approximately 183 h on a single NVIDIA RTX A6000 graphics card (48 GB GPU memory) running on a Linux system. The training process used a batch size of 8 and the Adam optimizer with a learning rate of 0.0001. Figure S5 shows the evolution of the training MAE (L1) loss and the validation peak signal-to-noise ratio (PSNR), where the loss rapidly converged to approximately 0.008 after 30 epochs, accompanied by similar convergence of the validation PSNR to approximately 27.5. These results demonstrate stable and robust training, and the PSNR value of 27.5 further confirms that the trained PocomAI-sup can effectively refine coarse voxel grids. Checkpoints of the trained PocomAI-sup were saved for the AI-driven framework PocomAI.

1.4 PocomAI-seg3D

PocomAI-seg3D adopts an architecture similar to that of PocomAI-seg, in which the 2D convolutional neural networks are replaced by their 3D counterparts. PocomAI-seg3D predicts the boundaries of each filler instance in a composite voxel grid, outputting an instance-segmented voxel grid in which each filler instance is assigned a distinct label.

PocomAI-seg3D was trained using fine voxel grids V_f and segmented voxel grids V_s from the multimodal dataset with a training-to-validation split ratio of 0.8:0.2. Training for 100 epochs required approximately 95 h on a single NVIDIA RTX A6000 graphics card (48 GB GPU memory) running on a Linux system. The training process used a batch size of 16 and the Adam optimizer with a learning rate of 0.0001. Figure S6 shows the evolution of the training loss and validation MAE, where the loss gradually converged to approximately 0.008 after 60 epochs, accompanied by similar convergence of the validation MAE. These results demonstrate stable and robust training, and the small validation MAE further confirms that the trained PocomAI-seg3D can accurately segment voxel grids. Checkpoints of the trained PocomAI-seg3D were saved for the AI-driven framework PocomAI.

1.5 PocomAI-ident

PocomAI-ident is a CNN-based model comprising four 3D convolutional layers, each followed by a 3D max-pooling layer, followed by a global average pooling layer and two fully connected layers with ReLU

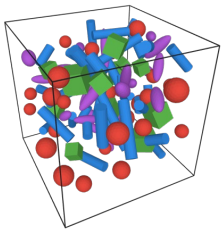
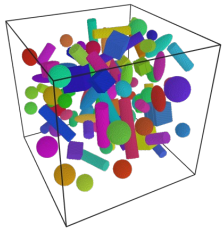
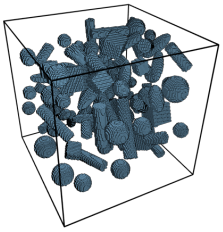
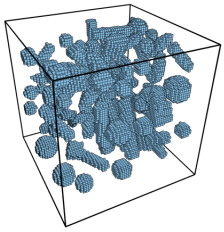
activation functions. The final fully connected layer, without activation, outputs the predicted filler type t, coordinate $\boldsymbol{c}$, size $\boldsymbol{s}$, and unit vector $\boldsymbol{v}$ of a given filler instance from a segmented voxel grid. The predicted filler parameters of all filler instances in a composite voxel grid are subsequently used to generate a mesh representation through CAD modeling.

PocomAI-ident was trained using segmented voxel grids V_s together with their corresponding filler information from the multimodal dataset with a training-to-validation split ratio of 0.8:0.2. Training for 100 epochs required approximately 53 h on a single NVIDIA RTX A6000 graphics card (48 GB GPU memory) running on a Linux system. The training process used a batch size of 16 and the Adam optimizer with a learning rate of 0.0001. Figure S7 shows the evolution of the training MAE loss and validation MAE, where the loss gradually converged and stabilized at approximately 0.01 after 80 epochs, accompanied by similar convergence of the validation MAE. These results demonstrate stable and robust training, and the small validation MAE further confirms that the trained PocomAI-ident can accurately predict filler parameters. Checkpoints of the trained PocomAI-ident were saved for the AI-driven framework PocomAI.

1.6 PocomAI-homo

PocomAI-homo is a CNN-based model comprising ten 3D convolutional layers, with a 3D max-pooling layer inserted after every two convolutional layers, followed by a global average pooling layer and three fully connected layers with ReLU activation functions. The final fully connected layer, without activation, outputs the predicted solid volume fraction ϕ, Young's modulus E, and thermal conductivity k of a composite voxel grid.

PocomAI-homo was trained using voxel grids V_f of size $128 \times 128 \times 128 \times 1$ labeled with their corresponding material properties from the multimodal dataset with a training-to-validation split ratio of 0.8:0.2. Training for 100 epochs required approximately 78 h on a single NVIDIA RTX A6000 graphics card (48 GB GPU memory) running on a Linux system. The training process used a batch size of 16 and the Adam optimizer with a learning rate of 0.0001. Figure S8 shows the evolution of the training MAE loss and validation MAE, where the loss gradually converged to approximately 0.02 after 30 epochs, accompanied by similar convergence of the validation MAE. These results demonstrate stable and robust training, and the small validation MAE further confirms that the trained PocomAI-homo can accurately predict the material properties of composite voxel grids. Checkpoints of the trained PocomAI-homo were saved for the AI-driven framework PocomAI.

Data	Mesh with filler information	Segmented voxel grid	Fine voxel grid	Coarse voxel grid
Visualization				
Symbol	$\left\{\begin{matrix} t,(c_1,c_2,c_3), \\ (s_1,s_2,s_3),(v_1,v_2,v_3) \end{matrix}\right\}^M$	V_s	V_f	V_c
Shape	M×1, M×3, M×3, M×3	128×128×128×M	128×128×128×1	64×64×64×1
[min, max]	[0, 1], [0, 1], [0, 1], [0, 1]	[0, 1]	[-1, 1]	[-1, 1]

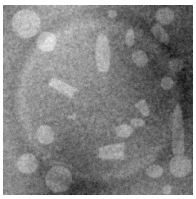
Data	Segmented binary images	X-ray CT-style images	Normalized material properties
Visualization			N/A
Symbol	$\{I_{s,i}\}_{i=1}^{M=256}$	$\{I_{CT,i}\}_{i=1}^{M=256}$	(ϕ, E, k)
Shape	M×256×256×1	M×256×256×1	64×64×64×1
[min, max]	[0, 1]	[0, 1]	[0, 1]

Figure S1: Multimodal data used for training PocomAI, with data visualization, symbol, array shape, and range.

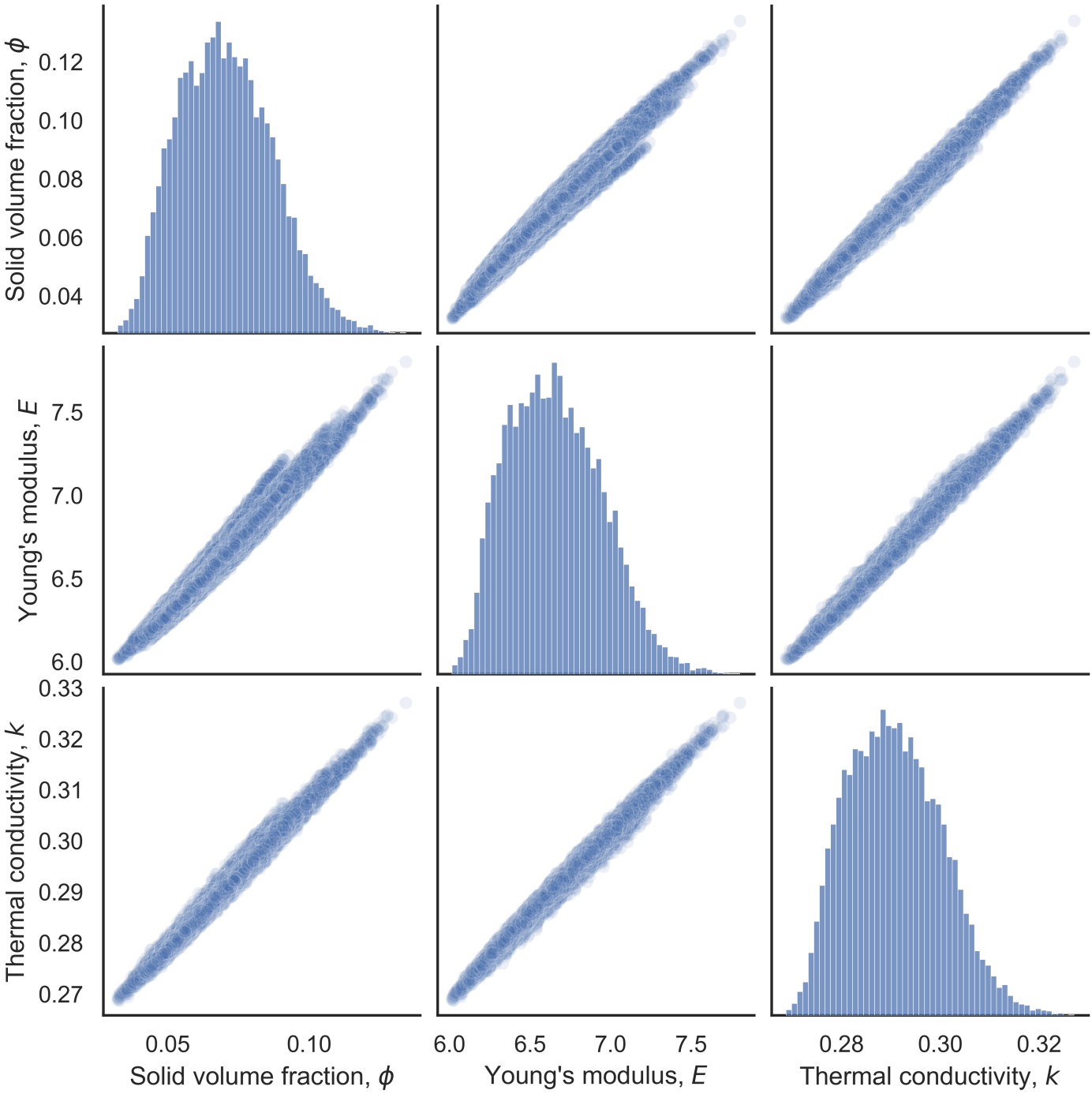


Figure S2: Dataspace of the material properties of prepared polymer composites in 2D projections.

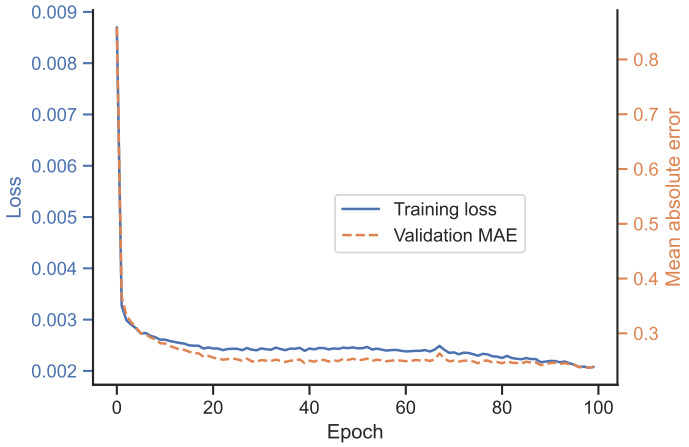


Figure S3: Training loss and validation mean absolute error of PocomAI-seg.

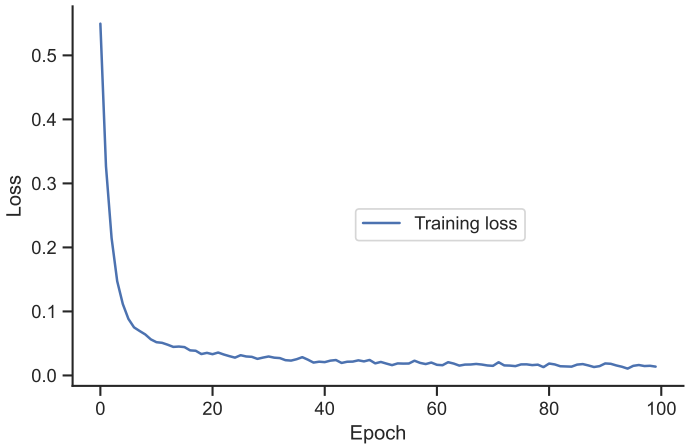


Figure S4: Training loss of PocomAI-gen.

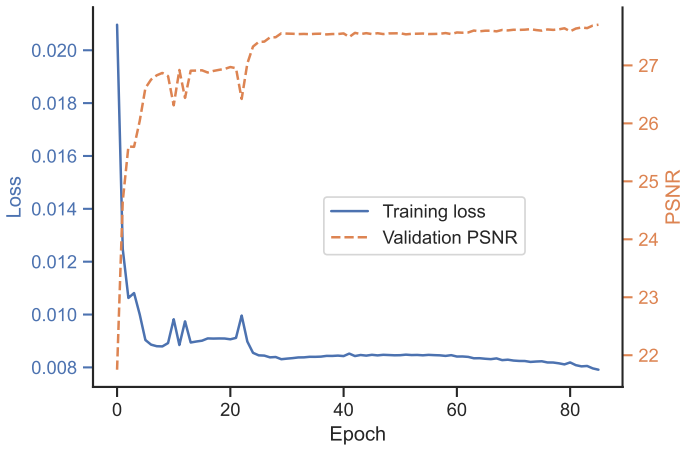


Figure S5: Training loss and validation peak signal to noise ratio (PSNR) of PocomAI-sup.

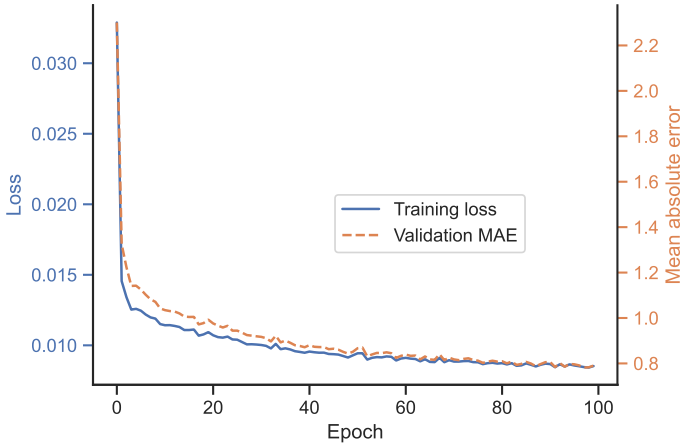


Figure S6: Training loss and validation mean absolute error of PocomAI-seg3D.

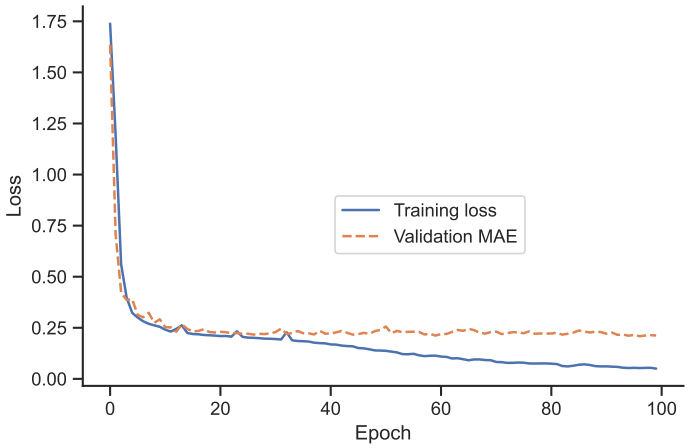


Figure S7: Training loss and validation mean absolute error of PocomAI-ident.

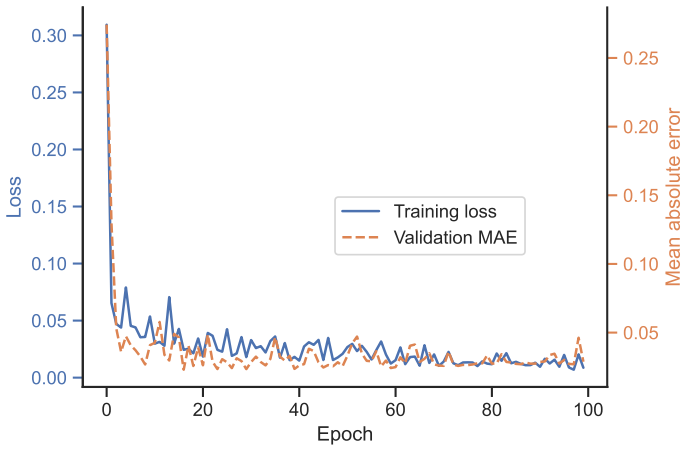


Figure S8: Training loss and validation mean absolute error of PocomAI-homo.

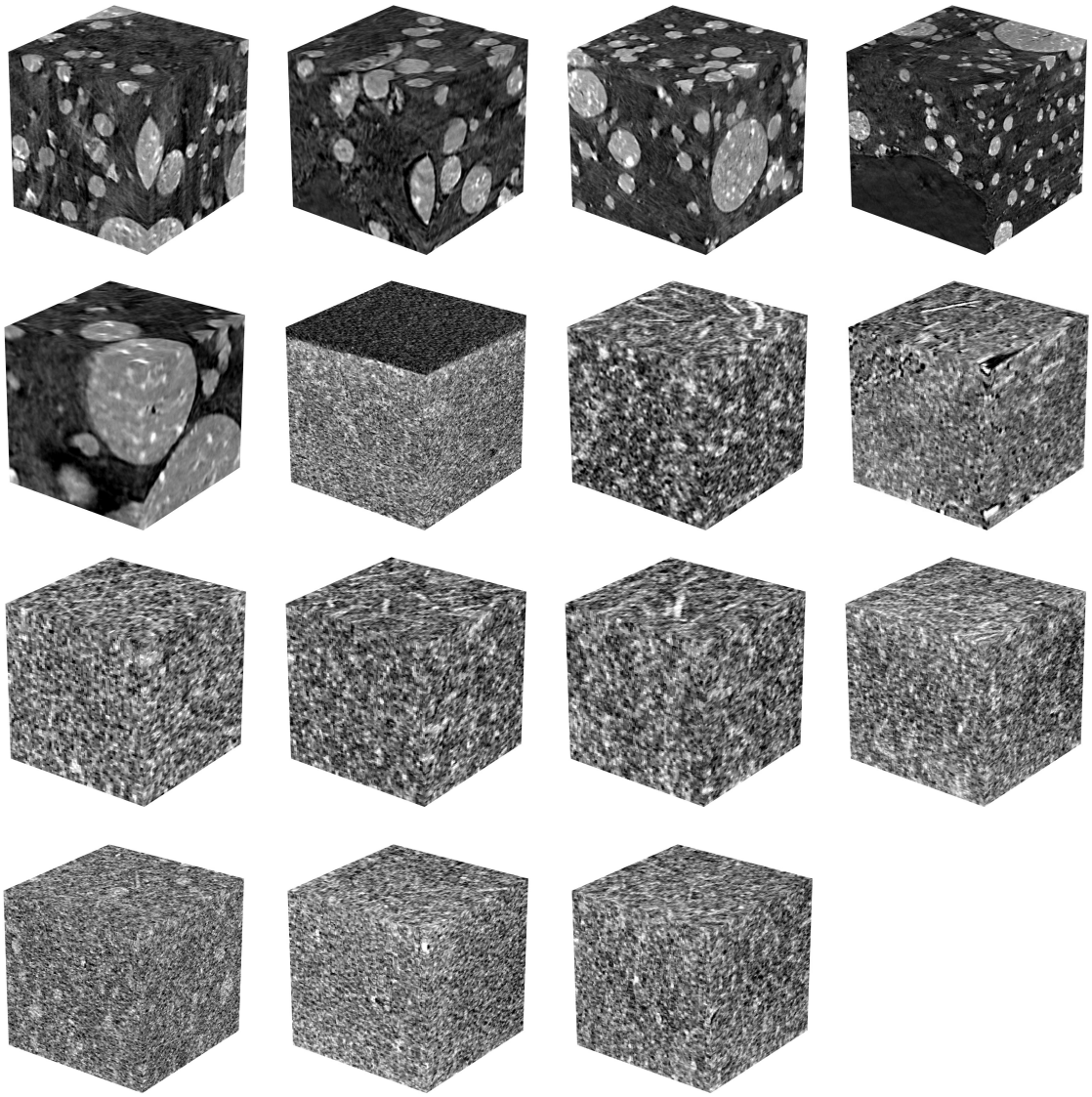


Figure S9: Voxel representation of stacked image slices of composites obtained from experimental X-ray. These composites, fabricated via compounding, contain spherical fillers, cylindrical fillers, or mixed sphere/cylinder fillers.

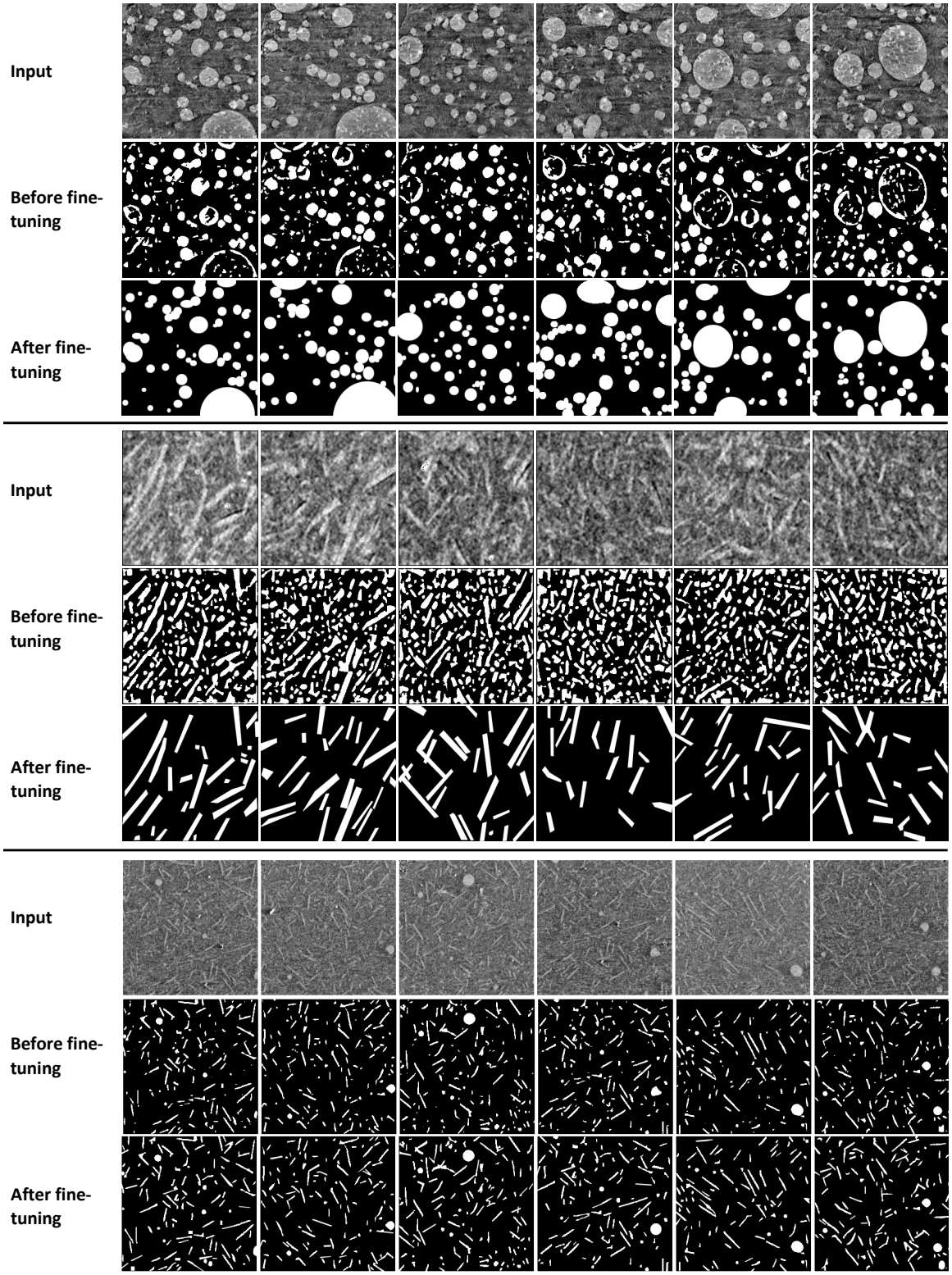


Figure S10: Experimental X-ray CT image segmentation using PocomAI-seg before and after fine-tuning.

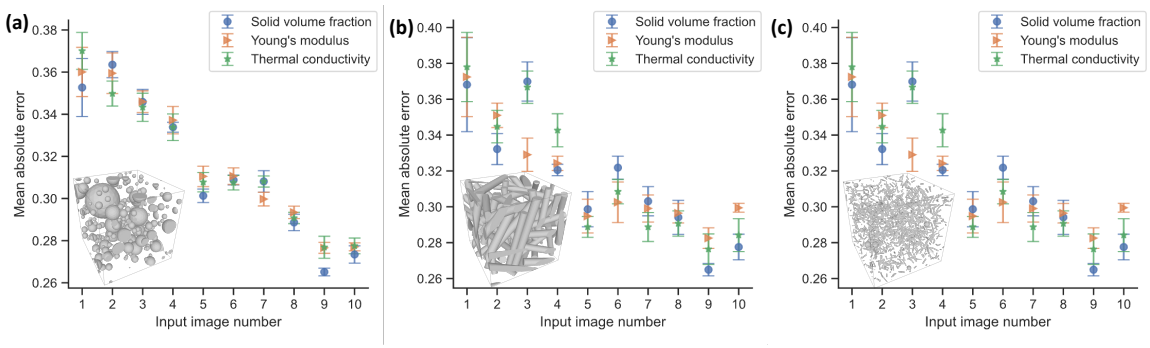


Figure S11: Statistical study for different sets of input images for the property-level reconstruction of experimentally obtained three composite classes via PocomAI. For each input set, a 3D was generated and its solid volume fraction, Young's modulus, and thermal conductivity were predicted using PocomAI-homo. The resulting property distributions were compared with those of the reference structures reconstructed from the complete X-ray CT datasets.

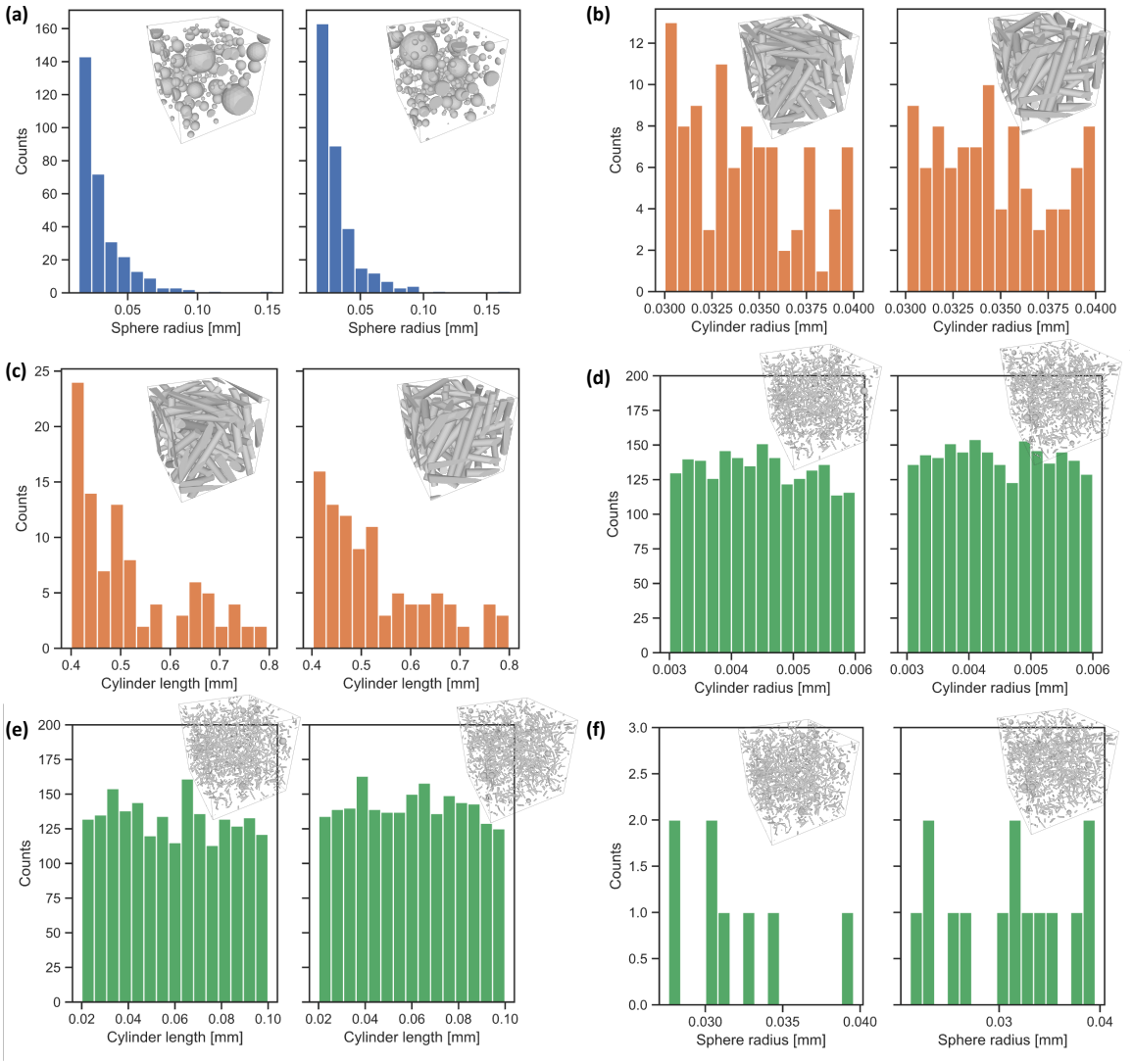


Figure S12: Statistical comparison of PocomAI-generated and Avizo-reconstructed structures in terms of their morphological fillers, in comparison to Figure 10. (a) Histograms of the sphere filler radius of PocomAI-generated (left) and reconstructed (right) spherical composites, along with histogram intersection score of 0.8593. (b) Histograms of the cylinder filler radius of PocomAI-generated (left) and reconstructed (right) cylindrical composites, along with histogram intersection score of 0.8316. (c) Histograms of the cylinder filler length of PocomAI-generated (left) and reconstructed (right) cylindrical composites, along with histogram intersection score of 0.8105. (d) Histograms of the cylinder filler radius of PocomAI-generated (left) and reconstructed (right) mixed sphere/cylinder composites, along with histogram intersection score of 0.9237. (e) Histograms of the cylinder filler length of PocomAI-generated (left) and reconstructed (right) mixed sphere/cylinder composites, along with histogram intersection score of 0.9274. (f) Histograms of the sphere filler radius of PocomAI-generated (left) and reconstructed (right) mixed sphere/cylinder composites, along with histogram intersection score of 0.3571. The histogram intersection score S_{HI} was calculated by $S_{\mathrm{HI}} = \sum_i \min(p_i, q_i)$, where where p_i and q_i are the probabilities in bin of histograms.

References

[1] Xuebin Qin, Deng-Ping Fan, Chenyang Huang, Cyril Diagne, Zichen Zhang, Adrià Cabeza Sant'Anna, Albert Suarez, Martin Jagersand, and Ling Shao. Boundary-aware segmentation network for mobile and web applications. *arXiv preprint arXiv:2101.04704*, 2021.

[2] Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition. In *Proceedings of the IEEE conference on computer vision and pattern recognition*, pages 770–778, 2016.

[3] Pieter-Tjerk De Boer, Dirk P Kroese, Shie Mannor, and Reuven Y Rubinstein. A tutorial on the cross-entropy method. *Annals of operations research*, 134(1):19–67, 2005.

[4] Zhou Wang, Eero P Simoncelli, and Alan C Bovik. Multiscale structural similarity for image quality assessment. In *The thrity-seventh asilomar conference on signals, systems & computers, 2003*, volume 2, pages 1398–1402. Ieee, 2003.

[5] Gellért Máttyus, Wenjie Luo, and Raquel Urtasun. Deeproadmapper: extracting road topology from aerial images. In *Proceedings of the IEEE international conference on computer vision*, pages 3438–3446, 2017.

[6] Jonathan Ho, Ajay Jain, and Pieter Abbeel. Denoising diffusion probabilistic models. *Advances in neural information processing systems*, 33:6840–6851, 2020.

[7] Xiaoyang Zheng, Junichiro Shiomi, and Takayuki Yamada. Optimizing metamaterial inverse design with 3d conditional diffusion model and data augmentation. *Advanced Materials Technologies*:2500293, 2025.

[8] David Foster. *Generative deep learning.* " O'Reilly Media, Inc.", 2022.

[9] Olaf Ronneberger, Philipp Fischer, and Thomas Brox. U-net: convolutional networks for biomedical image segmentation. In *Medical image computing and computer-assisted intervention–MICCAI 2015: 18th international conference, Munich, Germany, October 5-9, 2015, proceedings, part III 18*, pages 234–241. Springer, 2015.

[10] Bee Lim, Sanghyun Son, Heewon Kim, Seungjun Nah, and Kyoung Mu Lee. Enhanced deep residual networks for single image super-resolution. In *Proceedings of the IEEE conference on computer vision and pattern recognition workshops*, pages 136–144, 2017.